ASSIGNMENT

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Redux

→ Redux is an open source Java Script library used for state management. It was commonly used with React but can be integrated with any JS app.

→ In large applications, passing state between many components through props becomes complex and hard to manage. Redux solves this by storing the entire application state in a single global object called the 'STORE'.

→ STATE is read only and the only way to update state is by dispatching an action

Store - The store is the central object that hold the entire state of an application. It is created using the create Store (reducer) function. It has 3 important methods:

Get State() - returns the current state
dispatch(action) - sends an action to the reducer
Subscribe(listener) -

Action - An action is a plain Javascript object that describes an event that occured in the application. Every action must have a 'type' property, which is a string describing what happened.

Reducer - A reducer is a pure function that takes two arguments - the current state and action. and returns a new state. It contains the logic that and decides what it should return

Redux Flow

User Clicks Button → Action Dispatched → Reducer Runs → State Updates → UI re Renders

CODE:

src/store.js

```
import { createStore} from 'redux';

function reducers( state = { count: 0}, action) {
  switch (action.type) {
    case 'Increment': return { count: state.count +1};
    case 'Decrement': return {count: state.count - 1};
    default: return state;
  }
}

export const store = createStore(reducer);
```

src/counter.jsx

```
import { useSelector, useDispatch} from 'react-redux';

export default function Counter () {
  const count = useSelector( state => state.count);
  const dispatch= useDispatch();

return (
  <div style= { { textAlign: 'center', margin: '100'}}>
    <h1> {count}</h1>
  <button onClick={() => dispatch({ type: 'Increment'})}> +</button>
<button onClick={()=> dispatch({ type='Decrement'})}> - </button>
</div> ); }
```

src/App.jsx

```
import {Provider} from 'react-redux';
import {store} from './store';
import {counter from './counter';

export default function App () {

return (
  Provider store = {store}>
    <counter />
    </Provider>

  );
}
```